



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization experienced a DoS attack caused by a flood of ICMP packets that overwhelmed the network through an unconfigured firewall. The incident disrupted internal services for two hours until ICMP traffic was blocked and critical systems were restored. A post-incident investigation revealed firewall misconfiguration and lack of monitoring as key weaknesses. To reduce future risk, the security team implemented ICMP rate limiting, source IP validation, IDS and IPS filtering, and network monitoring. Using the NIST Cybersecurity Framework, the organization will improve its ability to identify vulnerabilities, protect assets, detect abnormal activity, respond effectively to incidents, and recover operations more efficiently.
Identify	Conduct regular vulnerability assessments and configuration audits on firewalls, routers, and network devices to identify gaps such as open services or misconfigured rules. Review asset inventories, network diagrams, and access privileges. Perform risk assessments to evaluate DoS exposure and business impact. Document findings in a risk register and align remediation priorities with organizational objectives and regulatory requirements.
Protect	Strengthen network protections by enforcing firewall configuration standards, implementing ICMP rate limiting, and validating source IP addresses to reduce

	spoofing risks. Establish security policies, change management procedures, and staff training on DoS awareness and incident reporting. Segment critical systems, apply least privilege, and maintain secure baselines. Ensure backups, redundancy, and secure configurations are documented and reviewed regularly.
Detect	Enhance monitoring through centralized logging, IDS and IPS alerts, and network traffic analysis to identify abnormal ICMP or volumetric patterns. Correlate events across systems using SIEM tools to improve visibility. Perform continuous monitoring and periodic tuning to reduce false positives and strengthen early detection of potential DoS activity.
Respond	Follow an established incident response plan to contain malicious traffic, prioritize critical services, and communicate with stakeholders. Use playbooks to isolate affected segments, block attack vectors, and validate system integrity. Preserve logs for forensic analysis, identify root causes, and document lessons learned. Update security controls, policies, and procedures to prevent recurrence and strengthen organizational response capability.
Recover	Restore affected systems to normal operation using validated backups and redundancy plans. Conduct post-incident reviews to evaluate business impact and recovery timelines. Implement improvements to resilience, such as failover testing and capacity planning. Incorporate findings into business continuity and disaster recovery strategies.

Reflections/Notes: